Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

For $a>0, t \in\left(0, \frac{\pi}{2}\right)$, let $x=\sqrt{a^{\sin ^{-1} t}}, y=\sqrt{a^{\cos ^{-1} t}}$. Then $1+\left(\frac{dy}{dx}\right)^2$ equals

(A) $\frac{x^2}{y^2}$

(B) $\frac{y^2}{x^2}$

(C) $\frac{x^2+y^2}{y^2}$

(D) $\frac{x^2+y^2}{x^2}$

Q.2

If $x=\sin^{-1} t$ and $y=\log\left(1-t^2\right)$; then $\left.\frac{\mathrm{d}^2 y}{\mathrm{d} x^2}\right|_{t=1/2}$ is

(A) $-8/3$

(B) $8/3$

(C) $3/4$

(D) $-3/4$

Q.3

If $\sqrt{\left(1-x^6\right)}+\sqrt{\left(1-y^6\right)}=a\left(x^3-y^3\right)$ and $\frac{dy}{dx}=f(x, y) \sqrt{\left(\frac{1-y^6}{1-x^6}\right)}$, then

(A) $f(x, y)=y / x$

(B) $f(x, y)=y^2 / x^2$

(C) $f(x, y)=2 y^2 / x^2$

(D) $f(x, y)=x^2 / y^2$

Q.4

If $x=f(t) \cos t-f'(t) \sin t$ and $y=f(t) \sin t+f'(t) \cos t$, then $\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2=$

(A) $f-f^n$

(B) $\left(f-f^2\right)^2$

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(C) $(f+f^n)^2$

(D) None of these

Q.5

Let $f(x)=4$ and $f'(x)=4$. Then $\lim_{x\to 2}\frac{xf(2)-2f(x)}{x-2}$ is given by

(A) 2

(B) -2

(C) -4

(D) 3

Q.6

Derivative of $\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2$ is

(A) $\frac{1}{2}$

(B) $1-\frac{1}{x^2}$

(C) 1

(D) $1+\frac{1}{x^2}$

Q.7

If f be a function given by $\mathrm{f(x)=2x^2+3x-5}$. Then, $f'(0)=mf'(-1)$, where m is equal to

(A) -1

(B) -2

(C) -3

(D) -4

Q.8

If $\frac{d}{dx}\left(\frac{1+x^4+x^8}{1+x^2+x^4}\right)=ax^3+bx$, then

(A) $a=4, b=2$

(B) $a=4, b=-2$

(C) $a=-2, b=4$

(D) None of these

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Q.9

If $f(x) = \left(\frac{\sin^m x}{\sin^n x}\right)^{m+n} \cdot \left(\frac{\sin^n x}{\sin^p x}\right)^{n+p} \cdot \left(\frac{\sin^p x}{\sin^m x}\right)^{p+m}$
then $f'(x)$ is equal to

(A) 0

(B) 1

(C) $\cos^{m+n+p} x$

(D) None of these

Q.10

If
$y = \frac{1}{1+x^{\beta-\alpha}+x^{\gamma-\alpha}} + \frac{1}{1+x^{\alpha-\beta}+x^{\gamma-\beta}} + \frac{1}{1+x^{\alpha-\gamma}+x^{\beta-\gamma}}$
then $\frac{dy}{dx}$ is equal to

(A) 0

(B) 1

(C) $(\alpha+\beta+\gamma)x^{\alpha+\beta+\gamma-1}$

(D) None of these

Q.11

Let $u(x)$ and $v(x)$ be differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{\sqrt{v(x)}}\right) = q$
,\r\nthen $\frac{p+q}{p-q}$ has the value equal to

(A) 1

(B) $\mathbf{0}$

(C) 7

(D) -7

Q.12

If $\sqrt{(1-x^{2n})} + \sqrt{(1-y^{2n})} = a\,(x^n - y^n)$, then $\sqrt{\left(\frac{1-x^{2\pi}}{1-y^{2n}}\right)}\frac{dy}{dx}$ is equal to

(A) $\frac{x^{n-1}}{y^{n-1}}$

(B) $\frac{y^{n-1}}{x^{n-1}}$

(C) $\frac{x}{y}$

(D) 1

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Q.13

If $x = \sec\theta - \cos\theta, y = \sec^{10}\theta - \cos^{10}\theta$ and $\left(x^2+4\right)\left(\frac{dy}{dx}\right)^2 = k\left(y^2+4\right)$, then k is equal to

(A) $\frac{1}{100}$

(B) 1

(C) 10

(D) 100

Q.14

The solution set of $f(x) > g(x)$ where $f(x) = (1/2)5^{2x+1}$ and $g(x) = 5^x + 4x\log_e 5$ is

(A) $(1, \infty)$

(B) $(0, 1)$

(C) $[0, \infty)$

(D) $(0, \infty)$

Q.15

Let $u(x)$ and $v(x)$ be differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{\mathrm{u(x)}}{\mathrm{v(x)}}\right)' = \mathrm{q}$, then $\frac{\mathrm{p+q}}{\mathrm{p-q}}$ has the value equal to

(A) 1

(B) 0

(C) 7

(D) -7

Answers & Solutions

Q.1 Answer:

$\frac{x^2+y^2}{x^2}$

Solution:

$\frac{dx}{dt} = \frac{1}{2}\left(a^{\sin^{-1}t}\right)^{-\frac{1}{2}} \cdot a^{\sin^{-1}t}\frac{1}{\sqrt{1-t^2}}\log a$

$= \frac{1}{2}\frac{\left(a^{\sin^{-1}t}\right)^{\frac{1}{2}}}{\sqrt{1-t^2}}\log a = \frac{1}{2}\frac{x}{\sqrt{1-t^2}}\log a$

$\frac{dy}{dt} = -\frac{1}{2}\left(a^{\cos^{-1}t}\right)^{-\frac{1}{2}} \cdot a^{\cos^{-1}t} \cdot \frac{1}{\sqrt{1-t^2}}\log a$

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$= -\frac{1}{2}\frac{y}{\sqrt{1-t^2}}\log a$

$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$ **so** $1 + \left(\frac{dy}{dx}\right)^2 = \frac{x^2+y^2}{x^2}$.

Q.2 Answer:

$-8/3$

Solution:

$\frac{dx}{dt} = \frac{1}{\sqrt{1-t^2}}, \frac{dy}{dt} = -\frac{2t}{1-t^2} \Rightarrow \frac{dy}{dx} = \frac{-2t}{\sqrt{1-t^2}}$

$\frac{d^2y}{dx^2} = -2\frac{d}{dt}\left(\frac{t}{\sqrt{1-t^2}}\right)\frac{dt}{dx}$

$= -2\frac{\left(\sqrt{1-t^2}+\frac{t^2}{\sqrt{1-t^2}}\right)\times\sqrt{1-t^2}}{(1-t^2)}$

$= -2\frac{1}{1-t^2}$

$\Rightarrow \left.\frac{d^2y}{dx^2}\right|_{t=1/2} = -2\cdot\frac{4}{3} = -\frac{8}{3}$

Q.3 Answer:

$f(x,y) = x^2/y^2$

Solution:

Put $x^3 = \sin\theta, y^3 = \sin\phi$, **then**

$(\cos\theta + \cos\phi) = a(\sin\theta - \sin\phi)$

$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\left(\frac{\theta-\phi}{2}\right) = 2a\cos\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$

$\Rightarrow \cot\left(\frac{\theta-\phi}{2}\right) = a$

$\Rightarrow \frac{\theta-\phi}{2} = \cot^{-1}a \Rightarrow \sin^{-1}x^3 - \sin^{-1}y^3 = 2\cot^{-1}a$

$\therefore \frac{3x^2}{\sqrt{(1-x^6)}} - \frac{3y^2}{\sqrt{(1-y^6)}}\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx} = \frac{x^2}{y^2}\sqrt{\left(\frac{1-y^6}{1-x^6}\right)}$

$\therefore f(x,y) = \frac{x^2}{y^2}$

Q.4 Answer:

$(f+f^n)^2$

Solution:

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$x = f(t)\cos t - f'(t)\sin t$

$y = f(t)\sin t + f'(t)\cos t$

$\frac{dx}{dt} = f'(t)\cos t - f(t)\sin t - f''(t)\sin t - f'(t)\cos t$

$= -f(t)\sin t - f''(t)\sin t$

$= -\sin\left[f(t) + f''(t)\right]$

$\frac{dx}{dt} = f'(t)\sin t - f(t)\cos t - f''(t)\sin t - f'(t)\sin t$

$= -f(t)\,\text{cost} - f''(t)\cos t$

$= \text{cost}\left[f(t) + f''(t)\right]$

$\left(\frac{dy}{dt}\right)^2 = \left\{\cos t\left[f(t) + f''(t)\right]\right\}^2$

$= (\cos t)^2\left\{f(t) + f''(t)\right]^2$

$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (\sin t)^2\left\{f(t) + f''(t)\right\}^2 + (\cos t)^2\left\{f(t) + f''(t)\right\}^2$

$= \left\{f(t) + f''(t)\right\}2$

Q.5 Answer:

-4

Solution:

We have, $\lim_{x\to 2}\frac{xf(2) - 2f(x)}{x-2}\left(\frac{0}{0}\right)$

By applying 'L' Hopital rule, we get

$= \lim_{x\to 2} f(2) - 2f'(x) = f(2) - 2f'(2)$

$= 4 - 2\times 4 = -4$

Q.6 Answer:

$1 - \frac{1}{x^2}$

Solution:

We have, $\frac{\mathrm{d}}{\mathrm{dx}}\left\{\left(\sqrt{\mathrm{x}} + \frac{1}{\sqrt{\mathrm{x}}}\right)^2\right\}$

$= \frac{d}{dx}\left\{x + \frac{1}{x} + 2\right\}$

$= \frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{x}) + \frac{\mathrm{d}}{\mathrm{dx}}\left(\mathrm{x}^{-1}\right) + \frac{\mathrm{d}}{\mathrm{dx}}(2) = 1 + (-1)\mathrm{x}^{-2} + 0$

$= 1 - \frac{1}{x^2}$

Q.7 Answer:

-3

Solution:

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We first find the derivatives of $f(x)$ at $x=-1$ and at x $=0$. We have, $f'(-1)=$
$\lim_{h\to 0}\frac{f(-1+h)-f(-1)}{h}$
$\lim_{h\to 0}\frac{\left[2(-1+h)^2+3(-1+h)-5\right]-\left[2(-1)^2+3(-1)-5\right]}{h}$
$=\lim_{h\to 0}\frac{2h^2-h}{h}=\lim_{h\to 0}(2h-1)=2(0)-1=-1$
and $f'(0)=\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$
$=\lim_{h\to 0}\frac{\left[2(0+h)^2+3(0+h)-5\right]-\left[2(0)^2+3(0)-5\right]}{h}$
$=\lim_{h\to 0}\frac{2h^2+3h}{h}=\lim_{h\to 0}(2h+3)=2(0)+3=3$
Clearly, $f'(0)=-3f'(-1)$

Q.8 Answer:

$a=4, b=-2$

Solution:

We have, $\frac{d}{dx}\left[\frac{\left(1+x^2+x^4\right)\left(1-x^2+x^4\right)}{\left(1+x^2+x^4\right)}\right]=ax^3+bx$
$\Rightarrow \frac{d}{dx}\left(1-x^2+x^4\right)=ax^3+bx$
$\Rightarrow -2x+4x^3=ax^3+bx$
$\Rightarrow a=4$ **and** $b=-2.$

Q.9 Answer:

0

Solution:

We have, $f(x)=\left(\sin^{m-n}x\right)^{m+n}\cdot\left(\sin^{n-p}x\right)^{n+p}\cdot\left(\sin^{p-m}x\right)^{p+m}$
$=\sin^{m^2-n^2}x\cdot\sin^{n^2-p^2}x\cdot\sin^{p^2-m^2}x$
$=(\sin x)^{m^2-n^2+n^2-p^2+p^2-m^2}=(\sin x)^0=1$
$\therefore f'(x)=0.$

Q.10 Answer:

0

Solution:

We have,
$y=\frac{1}{1+\frac{x^\beta}{x^\alpha}+\frac{x^\gamma}{x^\alpha}}+\frac{1}{1+\frac{x^\alpha}{x^\beta}+\frac{x^\gamma}{x^\beta}}+\frac{1}{1+\frac{x^\alpha}{x^\gamma}+\frac{x^\beta}{x^\gamma}}$
$=\frac{x^\alpha}{x^\alpha+x^\beta+x^\gamma}+\frac{x^\beta}{x^\alpha+x^\beta+x^\gamma}+\frac{x^\gamma}{x^\alpha+x^\beta+x^\gamma}$
$=\frac{x^\alpha+x^\beta+x^\gamma}{x^\alpha+x^\beta+x^\gamma}=1 \quad \therefore \frac{dy}{dx}=0$

Q.11 Answer:

1

Solution:

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$u(x) = 7v(x) \Rightarrow u'(x) = 7v'(x) \Rightarrow p = 7$

(given)

Again $\frac{u(x)}{v(x)} = 7 \Rightarrow \left(\frac{u(x)}{\sqrt{v(x)}}\right) = 0 \Rightarrow q = 0$

Now $\frac{p+q}{p-q} = \frac{7+0}{7-0} = 1$

Q.12 Answer:

$\frac{x^{n-1}}{y^{n-1}}$

Solution:

Put $x^n = \sin\theta$ **and** $y^n = \sin\phi$

Then, $(\cos\theta + \cos\phi) = a(\sin\theta - \sin\phi)$

$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right)$

$= 2a\cos\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$

$\Rightarrow \cot\left(\frac{\theta-\phi}{2}\right) = a$

$\Rightarrow \left(\frac{\theta-\phi}{2}\right) = \cot^{-1} a \Rightarrow \theta - \phi = 2\cot^{-1} a$

or $\sin^{-1} x^n - \sin^{-1} y^n = 2\cot^{-1} a$

Differentiating both sides, we get

$\frac{nx^{n-1}}{\sqrt{(1-x^{2n})}} - \frac{ny^{n-1}}{\sqrt{(1-y^{2n})}}\frac{dy}{dx} = 0$

$\therefore \quad \sqrt{\left(\frac{1-x^{2n}}{1-y^{2n}}\right)}\frac{dy}{dx} = \frac{x^{n-1}}{y^{n-1}}$

Q.13 Answer:

100

Solution:

$\because x^2 + 4 = (\sec\theta - \cos\theta)^2 + 4$

$= (\sec\theta + \cos\theta)^2$

Similarly, $y^2 + 4 = \left(\sec^{10}\theta + \cos^{10}\theta\right)^2 \ldots$ **(ii)**

Now, $\frac{dy}{d\theta} = \sec\theta\tan\theta + \sin\theta = \tan\theta(\sec\theta + \cos\theta)$

and $\frac{dy}{d\theta} = 10\sec^9\theta\sec\theta\tan\theta - 10\cos^9\theta(-\sin\theta)$

$= 10\tan\theta\left(\sec^{10}\theta + \cos^{10}\theta\right)$

$\Rightarrow \quad \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{10\tan\theta\left(\sec^{10}\theta+\cos^{10}\theta\right)}{\tan\theta(\sec\theta+\cos\theta)}$

$\therefore \left(\frac{dy}{dx}\right)^2 = 100\frac{\left(\sec^{10}\theta+\cos^{10}\theta\right)^2}{(\sec\theta+\cos\theta)} = \frac{100\left(y^2+4\right)}{(x^2+4)}$

[from Eq. (i), (ii)]

or $\left(x^2+4\right)\left(\frac{dy}{dx}\right)^2 = 100\left(y^2+4\right)$

$\therefore \quad k = 100$

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Q.14 Answer:

$(0, \infty)$

Solution:

$\left(\frac{1}{2}\right) 5^{2x+1} \log 5 \times 2 > 5^x \log_e 5 + 4 \log_e 5$

$\Rightarrow \quad 5^{2x} \cdot 5 > 5^x + 4$

$\Rightarrow \quad 5 \cdot 5^{2x} - 5^x - 4 > 0$

$(5^x - 1)(5 \cdot 5^x + 4) > 0$

$\therefore 5^x > 1$

$x > 0 \Rightarrow x \in (0, \infty)$

Q.15 Answer:

1

Solution:

$\mathrm{u(x) = 7v(x) \Rightarrow u'(x) = 7v'(x) \Rightarrow p = 7}$ **(given)**

Again $\frac{\mathrm{u(x)}}{\mathrm{v(x)}} = 7 \Rightarrow \left(\frac{\mathrm{u(x)}}{\mathrm{v(x)}}\right)' = 0 \Rightarrow \mathrm{q} = 0$

Now $\frac{p+q}{p-q} = \frac{7+0}{7-0} = 1$

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